No.
Date

* lab7:

6, $X = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}$, $W_Q = W_K = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$, $W_V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$.

$Q = X.W_Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{pmatrix}$, $K = X.W_K = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{pmatrix}$, $V = X.W_V = \begin{pmatrix} 2 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$

$S = \frac{Q.K^T}{\sqrt{2}} = \begin{pmatrix} 2.8284 & 0 & 1.4142 \\ 0 & 2.8284 & 1.4142 \\ 1.4142 & 1.4142 & 1.4142 \end{pmatrix}$

$K^T = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \end{pmatrix}$

$A_{ij} = \frac{e^{S_{ij}}}{\sum_i e^{S_{ij}}} \Rightarrow A = \begin{pmatrix} 0.7679 & 0.0454 & 0.1867 \\ 0.0454 & 0.7679 & 0.1867 \\ 0.3333 & 0.3333 & 0.3333 \end{pmatrix}$

$Z = A.V = \begin{pmatrix} 1.7225 & 1 \\ 0.1775 & 1 \\ 1 & 1 \end{pmatrix}$

7, $X = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}$, $W_{Q1} = W_{K1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$.

$W_{V1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$

tương tự như trên head1 $= A_1.V_1 = \begin{pmatrix} 1.7225 & 1 \\ 0.2275 & 1 \\ 1 & 1 \end{pmatrix}$

$W_{Q2} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$, $W_{K2} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$, $W_{V2} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$.

TAKEYO
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No. _____

Date _____

$Q_2 = x \cdot W_{Q2} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$, $K_2 = x \cdot W_{K2} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 1 \end{pmatrix}$

$V_2 = x \cdot W_{V2} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$.

$S_2 = \frac{Q_2 \cdot K_2^T}{\sqrt{d_k}} = \begin{pmatrix} 3 & 4 & 4 \\ 2 & 3 & 3 \\ 3 & 4 & 4 \end{pmatrix} \Rightarrow S_2 = \frac{Q_2 \cdot K_2^T}{\sqrt{2}} = \begin{pmatrix} 2.1213 & 2.8284 & 2.8284 \\ 1.4142 & 2.1213 & 2.1213 \\ 2.1213 & 2.8284 & 2.8284 \end{pmatrix}$.

$A_2 = \text{Softmax}(S_2) \approx \begin{pmatrix} 0.1978 & 0.4011 & 0.4011 \\ 0.1978 & 0.4011 & 0.4011 \\ 0.1978 & 0.4011 & 0.4011 \end{pmatrix}$.

$\text{head}_2 \approx \begin{pmatrix} 1.4011 & 1.1978 \\ 1.4011 & 1.1978 \\ 1.4011 & 1.1978 \end{pmatrix}$

$C = \text{concat}(\text{head}_1, \text{head}_2)$

$\approx \begin{pmatrix} 1.7225 & 1 & 1.4011 & 1.1978 \\ 0.2775 & 1 & 1.4011 & 1.1978 \\ 1 & 1 & 1.4011 & 1.1978 \end{pmatrix}_{3\times4}$

$W^o = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}_{4\times4}$

$\text{multihead} \approx C \cdot W^o = \begin{pmatrix} 3.1236 & 2.1978 & 2.4011 & 2.9203 \\ 1.8736 & 2.1978 & 2.4011 & 2.4753 \\ 2.4011 & 2.1978 & 2.4011 & 2.1978 \end{pmatrix}$.

Cần concatenate khi áp dụng W_o vì concatenation giúp đầu ra của các attention heads thành 1 representation chung, giữ lại thông tin từng head $\xrightarrow{W_o}$ trọng số thích hợp thành representation mong muốn.

TAKEYO
タケヨ

No. _____

Date _____

10. pos = 0 ⇒ PE(0) = [0 1 0 1] = [sin 0 cos 0 sin 0 cos 0]

pos = 1 ⇒ PE(1) = (sin 1 cos 1 sin 0.01 cos 0.01)

= [0.8415 0.5403 0.0099 0.9999)

pos = 2 ⇒ PE(2) = [sin 2 cos 2 sin 0.02, cos 0.02]

≈ [0.9093 −0.4161 0.0199 0.9998]

11. $W_1 X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + b_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$h = \text{relu}(z) = \begin{bmatrix} \max(0,1) \\ \max(0,1) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$

$W_2 = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, b_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow y = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$